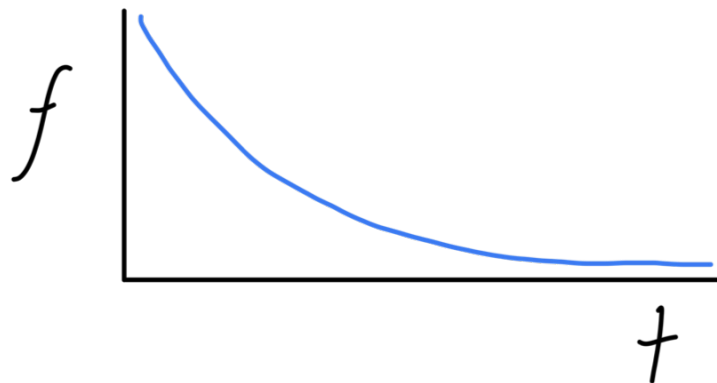


EXAMPLE OF $E(T)$ AND $V_{AR}(T)$

→ $T \sim \text{exponential}(\lambda)$

→ $f(t) = \lambda e^{-\lambda t} \quad t \geq 0$



MEAN: $E(T) = \int_0^{\infty} t f(t) dt =$

$$= \int_0^{\infty} \underbrace{t}_{u} \underbrace{\lambda e^{-\lambda t}}_{dv} dt =$$

$$u = t$$

$$v = -e^{-\lambda t}$$

$$du = dt$$

$$dv = \lambda e^{-\lambda t} dt$$

$$= \left[-t e^{-\lambda t} \right]_0^{\infty} + \int_0^{\infty} e^{-\lambda t} dt =$$

$$= 0 + \left[-\frac{1}{\lambda} e^{-\lambda t} \right]_0^{\infty} =$$

$$\frac{1}{\lambda}$$

VAR: $V_{AR}(T) = E(T^2) - (E(T))^2 =$

$$= \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

w/ $E(T^2) = \int_0^{\infty} t^2 \lambda e^{-\lambda t} dt = \dots = \frac{2}{\lambda^2}$